

NETWORKS AND COMPLEXITY

Solution 14-5

*This is an example solution from the forthcoming book *Networks and Complexity*.*

Find more exercises at <https://github.com/NC-Book/NCB>

Ex 14.5: Complex system step-by-step [Lvl. 3]

Let's go step-by-step through the analysis of the following system:

$$\begin{aligned}\dot{x} &= a + bxy \\ \dot{y} &= \frac{1}{2} - x\end{aligned}$$

- a) Compute the steady state of the system. Then find the Jacobian in the steady state, and compute the eigenvalues.

Solution

From the second equation we see

$$x^* = \frac{1}{2}$$

Substituting into the first equation yields the condition

$$0 = a + \frac{b}{2}y^* \quad (1)$$

and hence

$$y^* = -2\frac{a}{b} \quad (2)$$

Jacobian. We compute the elements of the Jacobian matrix.

$$J_{11} = \left. \frac{\partial \dot{x}}{\partial x} \right|_* = by^* = -2a \quad (3)$$

$$J_{12} = \left. \frac{\partial \dot{x}}{\partial y} \right|_* = bx^* = b/2 \quad (4)$$

$$J_{21} = \left. \frac{\partial \dot{y}}{\partial x} \right|_* = -1 \quad (5)$$

$$J_{22} = \left. \frac{\partial \dot{y}}{\partial y} \right|_* = 0 \quad (6)$$

(7)

hence

$$\mathbf{J} = \begin{pmatrix} -2a & b/2 \\ -1 & 0 \end{pmatrix} \quad (8)$$

Eigenvalues. We find the characteristic polynomial using the approach

$$\begin{pmatrix} -2a & b/2 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ B \end{pmatrix} = \lambda \begin{pmatrix} 1 \\ B \end{pmatrix}$$

which gives us the conditions

$$\begin{aligned} -2a + bB/2 &= \lambda \\ -1 &= \lambda B \end{aligned}$$

From the second condition we get $B = -1/\lambda$. Substituting this into the first condition yields

$$-2a - b/2\lambda = \lambda$$

which we can write as

$$\begin{aligned} 0 &= \lambda^2 + 2a\lambda + b/2 \\ 0 &= (\lambda + a)^2 - a^2 + b/2 \\ a^2 - b/2 &= (\lambda + a)^2 \\ \pm\sqrt{a^2 - b/2} &= \lambda + a \\ \lambda_{1,2} &= -a \pm \sqrt{a^2 - b/2} \end{aligned}$$

- b) Determine for which values of a and b the eigenvalues are complex.

Solution

The eigenvalues are complex when

$$a^2 - \frac{b}{2} < 0$$

which is the case when

$$2a^2 < b$$

- c) How does stability depend on a and b if the eigenvalues are complex?

Solution

Since we know that the results of the square root are imaginary in this case, the real part is just $\text{Re}(\lambda) = -a$ and hence, the steady state is stable if

$$a > 0.$$

- d) How does stability depend on a and b if the eigenvalues are real?

Solution

Because we are now assuming that the eigenvalues are real the square has real values and the leading eigenvalue is

$$\lambda_1 = -a + \sqrt{a^2 - b/2}.$$

From this equation we can see that there is no hope for stability if $a < 0$ as this would make the first term positive and the square root will be certainly positive. Hence $a > 0$ is a necessary condition for stability.

This eigenvalue is negative if

$$\begin{aligned} 0 &> -a + \sqrt{a^2 - b/2} \\ a &> \sqrt{a^2 - b/2} \\ a^2 &> a^2 - b/2 \\ 0 &< b/2 \\ 0 &< b \end{aligned}$$

Note that squaring both sides of an inequality can produce wrong results if the sides have different signs. However in this case we know the square root is positive and we have already determined that we need a to be positive. Hence if the eigenvalues are real the condition for stability is

$$a > 0, b > 0$$

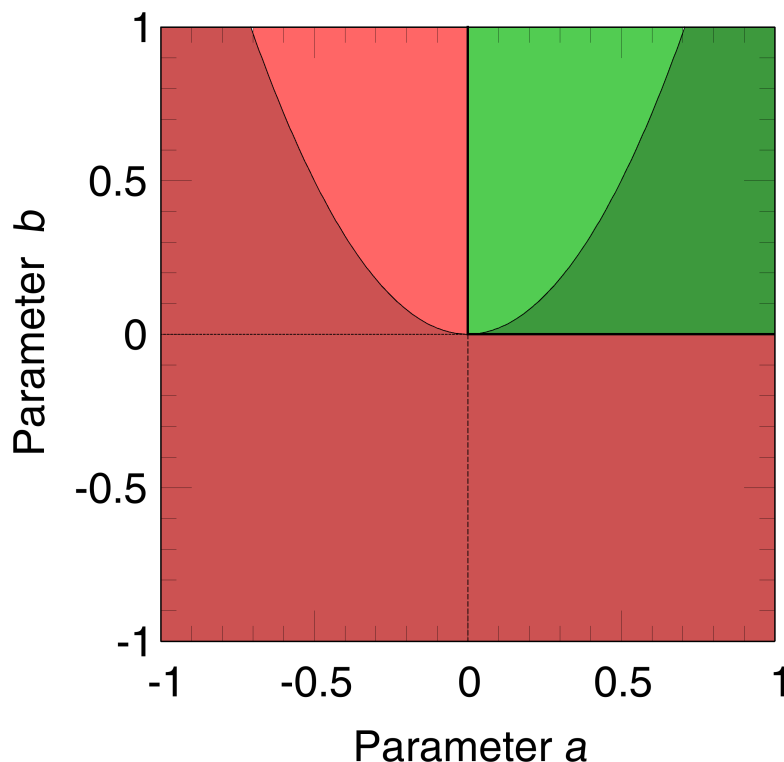
- e) Draw a diagram where a and b are the axis. Use lines to indicate where stability changes or eigenvalues become complex. Color in the area where the steady state is stable.

Solution

We know

- The eigenvalues are complex if $b > 2a^2$.
- If the eigenvalues are complex then the system is stable if $a > 0$.
- If the eigenvalues are real then the system is stable if $a > 0$ and $b > 0$.

We can graphically depict these cases follows:



The steady state is stable in the green (top-right) quadrant, which is bordered by the two stability conditions (strong lines). The brighter area (parabola) corresponds to the parameter region where the condition for complex eigenvalues is relevant.